Notes: Dewatripont and Maskin (RES, 1995)

Credit and Efficiency in Centralized and Decentralized Economies

Contents

1 Big picture

Question. How does the organization of credit markets affect project selection when creditors cannot tell, ex ante, whether a project is good or bad and cannot commit, ex post, not to refinance?

Setting. Entrepreneurs know more than creditors about project quality. A good project finishes quickly; a poor project takes longer. After first-period capital has already been sunk, creditors learn more and may want to continue even a bad project.

Core friction. Because sunk costs have already been incurred, a centralized creditor may rationally refinance a poor project ex post even though, ex ante, everyone would prefer that the threat of liquidation deter poor entrepreneurs from starting such projects in the first place.

Main theoretical contribution. The paper shows that *decentralized credit* can serve as a commitment device. When liquidity is fragmented across creditors and the initial lender's monitoring effort is not transferable to later lenders, refinancing becomes less attractive. That hardens the budget constraint.

Main baseline result. If a poor project is worth refinancing under centralization but not under decentralization, then decentralization screens poor projects out ex ante whereas centralization finances both good and poor projects.

Main endogenous-market-structure result. In the richer model with many investors and free entry, creditor size is determined endogenously. When commitment not to refinance is valuable, equilibrium credit is fragmented into small creditors. When all creditor sizes would refinance anyway, only large creditors survive because they monitor better.

Main short-termism result. The same fragmentation that disciplines bad long projects can also choke off good long projects. Once the model is extended to include a *very profitable* two-period project, a decentralized equilibrium with only small creditors can become excessively short-term.

Multiplicity result. The economy can have two very different equilibria: a short-term equilibrium with small creditors and only quick projects, and a long-term equilibrium with big creditors and high-return long projects. The long-term equilibrium can Pareto-dominate the short-term one.

Applications stressed by the paper. The framework is meant to illuminate two issues:

- soft budget constraints in centralized economies;
- the contrast between more market-based "Anglo-Saxon" finance and more bank-centered "German-Japanese" finance.

Economic takeaway. Centralization creates liquidity and internalizes monitoring, but it also makes continuation of weak projects too tempting. Decentralization restores discipline, but at the cost of placing too much weight on short-term returns.

2 Model environment and key objects

2.1 Agents, timing, and information

The baseline model has three dates, one entrepreneur, and either one centralized creditor or two decentralized creditors.

The sequence is:

- at date 0, the entrepreneur seeks first-period financing;
- after first-period capital has been committed, creditors learn more about project quality;
- if the project is still unfinished, a refinancing decision is made for period 2.

Interpretation. The creditor cannot distinguish good from poor projects when the initial contract is written. The important information arrives only *after* one unit of capital is sunk, which is exactly why commitment is difficult.

2.2 Project types and production

A project is either *good* (g) or *poor* (p).

- A good project is completed after one period and yields observable return $R_g > 1$.
- A poor project requires two periods for completion.
- Any project requires one unit of capital per period.

If a poor project is refinanced and completed, its date-2 observable return is the random variable

$$\tilde{R}_p \in \{0, \bar{R}_p\}, \qquad \bar{R}_p > 0.$$

Let α denote the prior probability that the entrepreneur is good.

Interpretation. Project quality is really project *duration*. A bad project is one that drags on too long relative to what financiers would want to support.

2.3 Entrepreneur private benefits

Because the entrepreneur has no bargaining power in the baseline model, she does not retain observable cash flow. Her payoff comes from private benefits:

- E_g: private benefit from a good completed project;
- E_t: private benefit if a poor project is terminated after the first period;
- E_p: private benefit if a poor project is completed.

The paper assumes

$$E_p \geq E_t.$$

The main case of interest is

$$E_p > 0 > E_t.$$

Interpretation. A poor entrepreneur wants funding only if she expects continuation. Termination must be painful enough to deter her from entering in the first place.

2.4 Monitoring technology

If the project is poor, the initial creditor can monitor during period 1. Let $a \in [0, 1]$ denote monitoring effort.

Monitoring affects the probability that the poor project pays $\bar{R}_p$:

$$\Pr(\tilde{R}_p = \bar{R}_p) = a.$$

Monitoring cost is

$$\psi(a),$$

with the usual shape restrictions:

$$\psi'(a) > 0, \qquad \psi''(a) > 0, \qquad \psi(0) = \psi'(0) = 0, \qquad \psi'(1) = \infty.$$

Interpretation. Monitoring raises the expected continuation value of a poor project, but it is costly. This is what creates the tension between preserving incentives and preserving commitment.

2.5 Centralization versus decentralization

Centralization. There is one bank B with two units of capital. It can both provide the first unit and, if it wishes, refinance with the second unit later on.

Decentralization. There are two banks, B_1 and B_2, each with only one unit of capital. The entrepreneur initially borrows from B_1. If the project needs another unit, she must turn to B_2.

The crucial informational assumption is that B_1's monitoring activity is *not observable* to B_2.

Interpretation. Decentralization here is really fragmented liquidity plus non-transferable soft information. It is not just "many banks" in a superficial sense.

2.6 Efficiency benchmark

The paper focuses attention on the case where:

$$R_g + E_g > 1$$

so good projects are socially desirable, and

$$\Pi_p^* + E_p < 2$$

so poor projects are socially undesirable even under the centralized creditor's better monitoring technology.

Interpretation. The interesting normative case is where good projects should be financed and poor projects should not. Then the question is which credit structure gets closer to that benchmark.

2.7 Why sunk costs generate softness

Once the first unit of capital has been sunk, the relevant ex post question is whether the second unit is worth adding. If the gross expected financial return from continuing a poor project exceeds one, the creditor will want to refinance even though, ex ante, everyone might have wanted a hard threat of liquidation.

Interpretation. This is the model's version of the soft budget constraint: continuation is privately rational ex post, but that very fact undermines discipline ex ante.

3 Model and theoretical results (all equations + interpretation)

3.1 Centralized benchmark

Because the single bank has all the liquidity, it fully internalizes the return from monitoring and continuation.

If the entrepreneur is good, the bank can extract the observable return R_g, so the net payoffs are:

$$\text{Entrepreneur: } E_g, \qquad \text{Bank: } R_g - 1.$$

If the entrepreneur is poor and the project is terminated after one period, payoffs are:

$$\text{Entrepreneur: } E_t, \qquad \text{Bank: } -1.$$

If the entrepreneur is poor and the project is refinanced, the bank chooses effort

$$a^* \in \arg\max_{a\in[0,1]} \left\{ a\bar{R}_p - \psi(a) \right\}. \tag{1}$$

The first-order condition is

$$\bar{R}_p = \psi'(a^*). \tag{2}$$

Define the centralized gross expected financial return on a refinanced poor project as

$$\Pi_p^* = a^*\bar{R}_p - \psi(a^*). \tag{3}$$

Because two units of capital are used in total, the bank's net payoff from a poor project that is refinanced is

$$\Pi_p^* - 2. \tag{4}$$

Interpretation. Centralization gives the bank the strongest possible monitoring incentive. That raises the continuation value of poor projects and therefore makes soft budget constraints more likely.

3.2 Decentralized benchmark

With decentralized credit, B_1 lends the first unit but B_2 must provide the second unit if the project is to continue.

Let $\hat{a}$ denote B_2's expectation of B_1's monitoring effort. To induce B_2 to participate, B_2 must be promised repayment $1/\hat{a}$ in the state where $\tilde{R}_p = \bar{R}_p$.

Thus B_1 chooses effort to maximize

$$a\left(\bar{R}_p - \frac{1}{\hat{a}}\right) - \psi(a). \tag{5}$$

In equilibrium $\hat{a}$ must be correct, so equilibrium monitoring a^{**} satisfies

$$\bar{R}_p = \psi'(a^{**}) + \frac{1}{a^{**}}. \tag{6}$$

Define the decentralized gross expected financial return on a refinanced poor project as

$$\Pi_p^{**} = a^{**}\bar{R}_p - \psi(a^{**}). \tag{7}$$

Since the extra term $1/a^{**}$ appears in the first-order condition, we have

$$a^{**} < a^* \qquad \Rightarrow \qquad \Pi_p^{**} < \Pi_p^*. \tag{8}$$

Interpretation. Even when all bargaining power is assigned to the initial bank, decentralization weakens monitoring because some of the continuation surplus leaks to the later lender. This is the paper's key commitment mechanism.

3.3 Proposition 1: decentralization as a commitment device

Under the key assumption

$$E_p > 0 > E_t,$$

there is a unique equilibrium under either organizational form.

The paper's central result is:

$$\Pi_p^* > 1 > \Pi_p^{**} \tag{9}$$

is a necessary and sufficient condition for project selection to differ across centralized and decentralized credit.

If this inequality holds:

- under **centralization**, both good and poor projects are financed, and poor projects are refinanced;
- under **decentralization**, only good projects are financed.

The logic is simple:

- if $\Pi_p^* < 1$, even the centralized bank will not refinance, so poor entrepreneurs are deterred under both systems;
- if $\Pi_p^{**} > 1$, even decentralized lenders refinance, so poor entrepreneurs seek financing under both systems;
- only in the middle case does decentralization harden the budget constraint enough to matter.

Interpretation. Decentralization does not help *always*; it helps only when continuation is attractive enough for a liquid centralized lender but not attractive enough for fragmented decentralized lenders.

3.4 Alternative interpretations in Section 2b

The authors emphasize that their mechanism does not depend on one very specific information problem.

They note that the same qualitative result would survive if:

- the entrepreneur affected project duration through hidden effort rather than project type being fixed from the outset;

- the key non-transferability problem were collusion or adverse selection rather than hidden monitoring effort;
- a centralized authority maximized social surplus rather than profit.

In fact, if the centralized lender cared about overall social surplus when deciding whether to continue, refinancing would become even more likely, which would make centralization's softness more severe.

Interpretation. The commitment advantage of decentralization is meant to be a broad organizational insight, not a quirk of one exact contracting protocol.

3.5 Section 3: endogenous creditor size

The paper then moves from a one-entrepreneur/two-bank benchmark to a market with:

- a population of n entrepreneurs;
- an indefinitely large population of identical investors with small endowments;
- free entry into bank formation.

Investors can pool to form creditors. Because everybody is risk neutral, there is no independent diversification benefit, so the largest creditor that needs to form in equilibrium has two units of capital.

Hence the relevant creditor types are:

- **small creditors**: one unit of capital;
- **big creditors**: two units of capital.

A big creditor can either:

- finance one project and keep one unit liquid (*diversified*), or
- finance two projects and tie up all of its liquidity (*undiversified*).

The paper stresses that "size" is really *liquidity*, not total assets.

Interpretation. This is important conceptually. A soft budget constraint is a problem of too much readily deployable funding, not necessarily of sheer organizational scale.

3.6 Equilibrium logic with free entry

With many investors and free entry, equilibrium contracts are driven to break even.

This means organizational outcomes are pinned down by *composition effects*:

- which creditor type attracts poor entrepreneurs;
- which creditor type monitors better once it attracts them;
- whether a competing entrant can undercut a profitable contract without inheriting the bad pool.

Interpretation. The theory is not simply comparing two fixed institutions anymore. It is asking which institution would survive competitively once investors are allowed to reorganize themselves.

3.7 Propositions 2 and 3: when small creditors dominate

Suppose again that

$$\Pi_p^* > 1 > \Pi_p^{**}.$$

Proposition 2. There exists an equilibrium in which each of $n+1$ or more small creditors offers a first-period contract that just breaks even on good entrepreneurs. No big creditors offer first-period contracts.

Proposition 3. This equilibrium is essentially unique.

The proof logic is:

- only big creditors can attract poor entrepreneurs because only they can promise continuation;
- poor entrepreneurs are unprofitable when $\Pi_p^* < 2$;
- therefore any big-creditor contract that is attractive enough to receive some good entrepreneurs will also attract bad entrepreneurs and can be undercut by a small creditor that only serves good projects.

Interpretation. Once small creditors can screen by refusing to refinance, competition and free entry drive large creditors out. The hard budget constraint becomes an equilibrium market outcome, not just a mechanical comparison of two exogenous institutions.

3.8 Assumption A and Proposition 4: when big creditors dominate

Now consider the opposite case:

$$\Pi_p^{**} > 1.$$

Then both small and big creditors refinance poor projects, so commitment no longer separates them.

The paper introduces:

Assumption A. Slow entrepreneurs have an arbitrarily small probability of completing their projects in one period.

This prevents creditors from sharply improving their client pool by making tiny changes in terms.

Under Assumption A:

Proposition 4. There is a unique equilibrium in which:

- only big creditors are active;
- at least $n+1$ of them offer contracts that break even *on average* across good and poor projects;
- these contracts extract the entire observable return from poor projects.

Interpretation. If no bank size can avoid refinancing, then the efficient organizational form is the one with the strongest monitoring incentives, namely liquid big creditors.

3.9 Section 4: adding a very profitable long-run project

To show that decentralization can go too far, Section 4 introduces a third project:

- v: a two-period but very profitable project with deterministic payoff $R_v > 2$.

Good entrepreneurs can now choose between:

- a quick good project g;
- a slow but very profitable project v.

Poor entrepreneurs remain stuck with poor projects p.

Poor and very profitable projects are indistinguishable to creditors at the end of period 1. Let:

- E_v: private benefit if the very profitable project is completed;
- E_t: private benefit if it is terminated after one period.

The paper assumes $E_v \geq E_p$.

Interpretation. The discipline device that kills bad long projects can also kill good long projects because, at the refinancing stage, both look similar.

3.10 Beliefs and monitoring in the long-project extension

Let α' denote the creditor's belief that a two-period project is very profitable rather than poor.

A big creditor chooses monitoring effort $a^*(\alpha')$ to satisfy

$$(1 - \alpha')\bar{R}_p = \psi'(a^*(\alpha')) \,, \tag{10}$$

because monitoring matters only when the project is actually poor.

Similarly, a small creditor's effort is $a^{**}(\alpha')$, with the same logic as before but lower continuation incentives.

Interpretation. Pessimistic beliefs about long projects reduce monitoring and refinancing willingness, which in turn make long projects unattractive to good entrepreneurs. Beliefs and project choice therefore reinforce each other.

3.11 Proposition 5: short-term equilibrium

If

$$\Pi_p^{**} < 1,$$

there exists an equilibrium in which:

- only small creditors are active;
- all good entrepreneurs choose the one-period good project rather than the two-period very profitable project.

Why? Creditors believe that a project needing refinancing is likely to be poor. Under those beliefs, small creditors will not refinance two-period projects. Anticipating this, good entrepreneurs avoid long projects.

Interpretation. This is the paper's formal short-termism result. Fragmented credit creates discipline, but that very discipline can rule out socially attractive long-horizon investment.

3.12 Condition $(*)$ and Proposition 6: long-term equilibrium

The paper then shows that a better equilibrium may also exist.

Let α now denote the population share of good entrepreneurs. If all good entrepreneurs choose the very profitable project, a big creditor's break-even condition is governed by

$$\alpha R_v + (1-\alpha)a^*(\alpha)\bar{R}_p - \psi(a^*(\alpha)) - 2 > \alpha(R_g - 1). \tag{*}$$

Condition $(*)$ says that the average gross surplus from financing the pool of long projects exceeds what creditors could generate from funding only short good projects.

Under the additional assumption

$$E_v = E_g,$$

Proposition 6 states that if $(*)$ holds, there exists an equilibrium in which:

- only big creditors form;
- all good entrepreneurs choose the very profitable long project.

The proof constructs a contract $\hat{c}$ that pays:

- nothing to the entrepreneur if the project turns out poor;
- T_v if it is very profitable;
- T_g if it is good,

where

$$\alpha(R_v - T_v) + (1-\alpha)a^*(\alpha)\bar{R}_p - \psi(a^*(\alpha)) - 2 = 0 \tag{11}$$

and

$$R_g - 1 - T_g = 0. \tag{12}$$

Good entrepreneurs choose the very profitable project when

$$E_v + T_v > E_g + T_g, \tag{13}$$

which, under $E_v = E_g$, collapses exactly to condition $(*)$.

Interpretation. Optimistic beliefs about long projects can be self-fulfilling. Large creditors then provide the liquidity needed to sustain long-term investment.

3.13 Multiplicity and Pareto ranking

Propositions 5 and 6 imply two qualitatively distinct equilibria in the same economy:

- a **short-term equilibrium**: small creditors, quick projects;
- a **long-term equilibrium**: big creditors, very profitable long projects.

When $(*)$ holds, the long-term equilibrium Pareto-dominates the short-term one.

Interpretation. Organizational form and project horizon are jointly determined. A decentralized system can become stuck in an equilibrium that is disciplined but too myopic.

4 What makes the argument credible (assumptions, interpretation, and limits)

4.1 Why decentralization can commit not to refinance

The whole mechanism rests on a simple point: the initial lender's monitoring is valuable, but later lenders cannot fully observe or price it. Once liquidity is fragmented, the initial lender cannot capture the full marginal benefit from monitoring and so continuation is less attractive.

Interpretation. Decentralization works because it creates a wedge between ex post continuation value and the initial lender's private return from making continuation possible.

4.2 Why illiquidity both helps and hurts

Illiquidity helps by preventing bad projects from being rolled over too easily.

Illiquidity hurts because long-horizon good projects also need rollover. If two-period projects are lumped together at the refinancing stage, discipline against one class of long projects becomes hostility toward long projects in general.

Interpretation. The paper is not claiming that decentralization is always superior. It is showing a classic commitment–flexibility trade-off.

4.3 Why competition is essential in Section 3

Free entry does two things:

- it drives equilibrium contracts to break even;
- it allows the more efficient creditor type to undercut the less efficient type.

That is why the Section 3 results are sharper than the two-bank benchmark. The market itself selects the credit structure consistent with the underlying continuation technology.

Interpretation. The hard- or soft-budget outcome is not imposed; it is competitively sustained.

4.4 Why the key case is $E_p > 0 > E_t$

If $E_t > 0$, termination no longer deters poor entrepreneurs, so decentralization cannot screen them out.

If $E_p < 0$, poor entrepreneurs would not seek financing under either structure.

Thus the most interesting discipline problem arises exactly when poor entrepreneurs gain from continuation but lose from early liquidation.

Interpretation. The paper's screening result is not universal; it requires that entrepreneurs care enough about continuation.

4.5 Liquidity versus literal size

The authors explicitly note that "small" and "big" creditors should be thought of as *illiquid* and *liquid* creditors. What matters is how much immediately deployable capital a creditor controls once a project is in trouble.

Interpretation. This is why the model speaks naturally to soft budget constraints in centralized systems: a center that can always find another unit of finance behaves like an extremely liquid creditor.

4.6 What the model abstracts from

Important simplifications include:

- risk neutrality, so ordinary portfolio diversification plays no independent role;
- no collateral or secured lending;
- no legal penalties or reputational sanctions strong enough to deter poor projects on their own;
- only one key continuation decision rather than a long dynamic sequence;
- no detailed political economy of why a centralized authority might keep inefficient firms alive.

Interpretation. The paper is a clean theory of organizational commitment in credit markets, not a complete model of socialism, banking systems, or corporate governance.

4.7 Why the applications are plausible but stylized

For centrally planned economies, the paper offers an *economic* reason for soft budget constraints: continuation can be privately profitable for a liquid centralized lender.

For comparative finance, it offers an organizational explanation for why more decentralized systems may be more short-termist while more bank-centered systems may sustain long-term investment.

Interpretation. These are illuminating analogies, but the paper does not directly test them or build institution-specific detail into the model.

5 Figures and key theoretical results

5.1 Table 1: payoffs under centralization

Table 1: Payoffs under centralization

	Good project (assuming $E_g > 0$)	Poor project without refinancing	Poor project with refinancing
Entrepreneur	E_g	E_t	E_p
Bank	$R_g - 1$	-1	$\Pi_p^* - 2$

Read.

- A good project gives the bank a clean one-period payoff $R_g - 1$.
- A poor project that is stopped after one period costs the bank one unit and leaves the entrepreneur with E_t.
- A poor project that is continued yields bank payoff $\Pi_p^* - 2$, where the key object is the centralized gross continuation return Π_p^*.

Economic message. Centralization increases Π_p because the same lender monitors and refinances. That is exactly why soft budget constraints arise.

5.2 Table 2: payoffs under decentralization

Table 2: Payoffs under decentralization

	Good project (assuming $E_g > 0$)	Poor project without refinancing	Poor project with refinancing
Entrepreneur	E_g	E_t	E_p
B_1	$R_g - 1$	-1	$\Pi_p^{**} - 2$
B_2	0	0	0

Read.

- The entrepreneur's private benefits are unchanged.
- Relative to centralization, the important object is now Π_p^{**}, which is smaller because the first bank cannot internalize the full payoff from monitoring.
- The second bank just breaks even in the refinancing stage.

Economic message. Decentralization hardens the budget constraint not by changing the entrepreneur's preferences, but by reducing the initial creditor's incentive to make continuation attractive.

5.3 Regime map for Propositions 1–4

Table 3: Which creditor structure survives?

Condition	Active creditors	Project selection	Main message
$\Pi_p^* < 1$	Either structure behaves the same	Only good projects funded	Poor projects are not worth continuing even centrally
$\Pi_p^* > 1 > \Pi_p^{**}$	Small creditors dominate (Props. 2–3)	Only good projects funded in decentralized equilibrium	Fragmentation is valuable because it commits against continuation
$\Pi_p^{**} > 1$	Big creditors dominate (Prop. 4)	Good and poor projects both funded	Once commitment is impossible anyway, liquidity and monitoring efficiency win

Economic message. The paper's central comparative-static is not "decentralization is good"; it is "decentralization is useful precisely when it blocks continuation that centralization would allow."

5.4 Section 4 equilibrium comparison

Table 4: Short-term versus long-term equilibrium in Section 4

Equilibrium	Creditor structure	Good entrepreneur's choice	Welfare message
Proposition 5	Small creditors only	Short good project	Disciplined but potentially too myopic
Proposition 6	Big creditors only	Very profitable long project	Can Pareto-dominate the short-term equilibrium

Economic message. The model ends with a genuine coordination problem: the same economy can support a disciplined short-term outcome or a more efficient long-term outcome.

6 Short summary

- The paper studies how credit market organization affects continuation and project selection under asymmetric information.
- A centralized lender monitors more and therefore is more willing to refinance poor projects once sunk costs are in place.
- A decentralized credit structure fragments liquidity and reduces monitoring incentives, which makes the threat to terminate poor projects more credible.
- When $\Pi_p^* > 1 > \Pi_p^{**}$, decentralization finances only good projects while centralization finances both good and poor ones.
- In a competitive model with endogenous bank formation, small creditors survive exactly in the region where commitment is valuable.
- If all creditor sizes refinance poor projects anyway, large creditors dominate because they monitor better.
- After adding very profitable long-run projects, the same decentralization that creates discipline can also generate excessive short-termism.
- The economy may have two equilibria: a short-term/small-bank equilibrium and a long-term/big-bank equilibrium, with the latter potentially Pareto-superior.

7 Conclusion

7.1 Core conclusion

The paper shows that decentralized credit can improve efficiency by doing something a centralized lender cannot: making the threat *not* to refinance bad projects credible.

7.2 Economic intuition

Soft budget constraints arise because ex post continuation is often privately attractive once capital is sunk. Fragmenting liquidity and soft information across creditors weakens continuation incentives, which is bad for monitoring but good for commitment. That trade-off is the entire logic of the paper.

7.3 What the paper does and does not deliver

What it delivers is a clean organizational theory of financial discipline and short-termism.

What it does *not* deliver is an empirical test, a detailed institutional description of any particular country, or a full political-economy account of centralized financing systems. The applications to socialism and comparative finance are interpretive rather than directly estimated.

7.4 Takeaway

Too much centralization makes it too easy to throw good money after bad. Too much decentralization makes it too hard to fund slow but valuable projects. The efficient credit system balances commitment against continuation, and the balance can depend on equilibrium beliefs about project quality and horizon.